length of [E] as 1 immediately and the recursion stops.

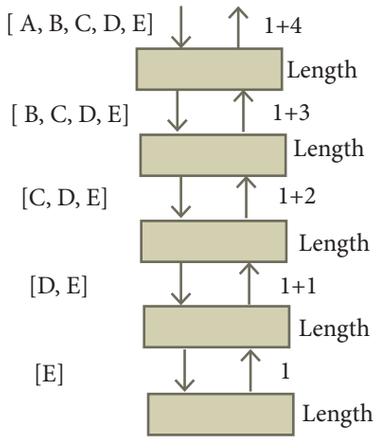

Figure 8.6: Recursive process with solvers and sub-solvers

The recursive process for length [A,B,C,D,E] is shown in Figure 8.7.

1	length [A,B,C,D,E]
2	= 1 + length [B,C,D,E]
3	= 1 +1 + length [C,D,E]
4	= 1 + 1+ 1 + length [D,E]
5	= 1 + 1+ 1 + 1 + length [E]
6	= 1 + 1 + 1 + 1 +1
7	= 1 +1 +1 + 2
8	= 1 + 1 + 3
9	= 1 + 4
10	= 5

Figure 8.7: Recursive process for computing the length of a sequence

8.4.2 Recursive problem solving

Each solver should test the size of the input. If the size is small enough, the solver should output the solution to the problem directly. If the size is not small enough, the solver should reduce the size of the input and call a sub-solver to solve the problem with the reduced input. For Example 8.10, solver's algorithm can be expressed as

$$\text{length of sequence} = \begin{cases} 1 & \text{if sequence has only one customer} \\ 1 + \text{length of tail,} & \text{otherwise} \end{cases}$$

To solve a problem recursively, the solver reduces the problem to sub-problems, and calls another instance of the solver, known as sub-solver, to solve the sub-problem. The input size to a sub-problem is smaller than the input size to the original problem. When the solver calls a sub-solver, it is known as recursive call. The magic of recursion allows the solver to assume that the sub-solver (recursive call) outputs the solution to the sub-problem. Then, from the solution to the sub-problem, the solver constructs the solution to the given problem.

As the sub-solvers go on reducing the problem into sub-problems of smaller sizes, eventually the sub-problem becomes small enough to be solved directly, without recursion. Therefore, a recursive solver has two cases:

1. **Base case:** The problem size is small enough to be solved directly. Output the solution. There must be at least one base case.
2. **Recursion step:** The problem size is not small enough. Deconstruct the problem into a sub-problem, strictly smaller in size than the given problem. Call a sub-solver to solve the sub-problem. Assume that the sub-solver outputs the solution to the sub-problem. Construct the solution to the given problem.

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